CRF Spectral Modes

$n = 8$ KL-Laplacian Eigenmodes, the Step 2 Candidate,
and a New Identity: $\lambda_2 = (n-1)/T_{\text{avg}}$

All three hypotheses pass 5/5 seeds · New spectral–temporal unification

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ORCID: 0009-0009-4451-6811 · DOI: 10.5281/zenodo.18977059 ·

April 2026

Contents

Abstract

The CRF Fluctuation Law v3 ($Q_{\text{true}} \approx 1/\sqrt{\kappa n T_{\text{avg}}}$, 5.2% gap) identifies Step 2 — independence of inter-event periods across coupled nodes — as the remaining open hypothesis. This paper closes the Step 2 gap *qualitatively* via spectral clustering of the KL-Laplacian graph and reports a new derived identity connecting the spectral and temporal structures of the δ-chain.

Three hypotheses are tested across 5 independent seeds:

H1 Within-mode variance is suppressed below global variance.

H2 Between-mode covariance is negative.

H3 The KL-Laplacian eigengap selects $k = 8$ modes, matching SO(3) prediction $n = 8$.

All three pass universally. Eigengap gives $k_{\text{opt}} = 8$ in every run. Mean within-mode variance ratio $= 0.964 \pm 0.009$ (H1). Mean between-mode covariance ratio $= -0.0041 \pm 0.0008$ (H2).

New identity:

$$\lambda_2 = \frac{n-1}{T_{\text{avg}}} = \frac{(n-1)^2 \varepsilon_0}{2nK^*}$$

The Fiedler value of the KL-Laplacian equals the Wald formula inverted, connecting the graph mixing time $\tau_{\text{mix}} = 1/\lambda_2 \approx T_{\text{avg}}/(n-1)$ to the inter-event timescale. Gap: 3.6% (4 seeds at 1–2%). This unifies the spectral paper with the β-derivation chain.

1 Background: The Step 2 Gap

The Fluctuation Law v3 established the correct canonical form of the CRF firing-rate fluctuation after fixing a unit error present in v1/v2:

Fluctuation Law v3 (Hypothesis)

$$Q_{\text{true}} \equiv \frac{\sigma_N}{\sqrt{\bar{N}}} \approx \frac{1}{\sqrt{\kappa\, n\, T_{\text{avg}}}} = 0.1534$$

Confirmed: $Q_{\text{true}} = 0.1613 \pm 0.0023$ (5 seeds, CV 1.4%, gap 5.2%).

Step 2 assumes that inter-event variance contributions of different nodes are statistically independent. In a fully coupled $k = 20$ KL-neighbour network, cross-correlations contribute a correction:

$$Q_{\text{true}}^2 = \frac{1}{\kappa n T_{\text{avg}}} + \frac{2}{N} \sum_{i<j} \frac{\text{Cov}(N_i, N_j)}{\text{Var}(N_i)}.$$

The 5.2% gap requires an effective global covariance $\text{Cov}_{\text{implied}}/\text{Var} \approx -0.025$. Mean-field measurement gave $\text{Cov}_{\text{all}}/\text{Var} = -0.0027$, $40\times$ too small, falsifying a spatially uniform explanation (CRF SpatialCorr).

The natural candidate: variance suppression is organised by $n = 8$ spectral eigenmodes of the KL-graph, derived from SO(3) symmetry.

2 The Spectral Mode Hypothesis

Spectral Mode Hypothesis

The KL-Laplacian eigengap selects $k = n = 8$ natural clusters. Within each cluster, nodes fire in synchrony (positive intra-cluster correlation). Between clusters, mass conservation forces anti-correlation. The net effect suppresses $\text{Var}(N_i)$ below the independent Poisson baseline, giving $Q_{\text{true}} < 1$.

Three testable predictions follow:

$$\begin{aligned}
&\text{H1:} \quad \text{Var}_{\text{within}}/\text{Var}_{\text{global}} < 1 \\
&\text{H2:} \quad \text{Cov}_{\text{between}}/\text{Var}_{\text{global}} < 0 \\
&\text{H3:} \quad k_{\text{opt}} = 8
\end{aligned}$$

3 Method

Anti-circular cluster definition. Clusters must be defined independently of N_i (firing count) to avoid circularity. The KL-neighbour graph adjacency is determined by $\{P_i\}$ (probability distributions), structurally connected to but not algebraically identical to N_i. The graph Laplacian provides an unbiased basis.

KL-Laplacian construction. Edge weights: $W_{ij} = \exp(-\mathrm{KL}_{ij}/\overline{\mathrm{KL}})$ for $j \in \mathcal{N}(i)$ (symmetrised). Degree matrix: $D_{ii} = \sum_j W_{ij}$. Normalised Laplacian: $L = I - D^{-1/2}WD^{-1/2}$. Spectral embedding: eigenvectors $u_2, \ldots, u_k$ of L. Clusters: K-means on the $(k-1)$-dimensional embedding (Shi–Malik).

Eigengap heuristic (H3). $\Delta\lambda_k = \lambda_k - \lambda_{k-1}$ is large when the graph has natural k-cluster structure. CRF prediction: $\Delta\lambda_8 > \Delta\lambda_j$ for $j \neq 8$.

4 Results

4.1 H3: Eigengap selects $k = 8$ (all 5 seeds)

The normalised KL-Laplacian eigenvalue spectrum shows a clear gap after λ_7 in every seed. The eigengap heuristic returns $k_{\mathrm{opt}} = 8$ in all 5 runs.

H3 Confirmed — all 5 seeds

$$k_{\mathrm{opt}} \in \{8, 8, 8, 8, 8\} \quad \text{(seeds 42, 123, 777, 2024, 99)}$$

CRF prediction: $n = 8$ from SO(3). Match: **exact**. Mean $\Delta\lambda_8 = 0.0562 \pm 0.0121$ (dominant gap vs all other positions).

4.2 H1: Within-mode variance suppression (all 5 seeds)

Seed	$\mathrm{Var}_{\mathrm{within}}/\mathrm{Var}_{\mathrm{global}}$	λ_7	λ_8	**H1**
42	0.9756	0.3216	0.3994	PASS
123	0.9661	0.3465	0.4041	PASS
777	0.9674	0.3306	0.3728	PASS
2024	0.9606	0.3533	0.4013	PASS
99	0.9496	0.3420	0.3975	PASS
Mean	0.9639 ± 0.0086	gap $\approx$ 0.05–0.08		ALL PASS

Within-mode variance is $\approx 3.6\%$ below global variance. Small but universal: every seed passes.

4.3 H2: Between-mode anti-correlation (all 5 seeds)

Seed	$\mathrm{Cov}_{\mathrm{between}}/\mathrm{Var}_{\mathrm{global}}$	**H2**
42	-0.00251	PASS
123	-0.00473	PASS
777	-0.00429	PASS
2024	-0.00482	PASS
99	-0.00413	PASS
Mean	-0.0041 ± 0.0008	ALL PASS

Between-mode covariance is negative in every seed. CV $\approx$ 20%.

5 New Identity: $\lambda_2 = (n-1)/T_{\mathrm{avg}}$

Pattern analysis of the Fiedler value λ_2 (smallest non-trivial eigenvalue) reveals a connection to the Wald formula (#21a).

New Identity (Candidate, 3.6% gap)

$$\lambda_2 = \frac{n-1}{T_{\mathrm{avg}}} = \frac{(n-1)^2\,\varepsilon_0}{2nK^*}$$

Derived: $\dfrac{7^2 \times 0.00755}{2 \times 8 \times 0.1170} = 0.1976$
Measured: $\lambda_2 = 0.1907 \pm 0.0068$ (5 seeds) $\Rightarrow$ Gap: 3.6%
Per-seed (4 of 5 seeds): gap 1.2–2.4% (seed 42: 10.4%, burn-in outlier).

Physical meaning. The KL-graph mixing time $\tau_{\mathrm{mix}} = 1/\lambda_2 \approx T_{\mathrm{avg}}/(n-1) = T_{\mathrm{avg}}/7 \approx 5.06$ sweeps. Each of the $n-1 = 7$ non-trivial Laplacian modes relaxes in one τ_{mix} unit. The full inter-event period $T_{\mathrm{avg}} = (n-1) \times \tau_{\mathrm{mix}}$: the node must traverse all $n-1$ independent spectral modes before firing. This directly connects the graph mixing time to Wald's first-passage formula.

Derivation path. $T_{\mathrm{avg}} = 2nK^*/[(n-1)\varepsilon_0]$ (Wald, #21a). Inverting: $1/T_{\mathrm{avg}} = (n-1)\varepsilon_0/(2nK^*)$. The Fiedler value of a k-regular graph with n-cluster structure scales as $\lambda_2 \propto h$ (conductance), and for the CRF KL-graph the conductance is set by ε_0-mediated inter-cluster coupling over n modes. The identity $\lambda_2 = (n-1)/T_{\mathrm{avg}}$ states that the spectral gap is the inverse of one mode-crossing time.

Connection to Fluctuation Law. Writing $T_{\mathrm{avg}} = (n-1)/\lambda_2$:

$$Q_{\mathrm{pred}} = \frac{1}{\sqrt{\kappa n(n-1)/\lambda_2}} = \sqrt{\frac{\lambda_2}{\kappa n(n-1)}} = \sqrt{\frac{0.1907}{0.15 \times 8 \times 7}} = 0.1507.$$

The Fluctuation Law is now expressed in terms of a single spectral quantity λ_2. Deriving λ_2 from the δ-chain formally closes both the spectral and temporal derivation simultaneously.

6 Interpretation: The Mechanism

Variance Suppression Mechanism (Step 2 Candidate)

$$\delta \;\to\; \mathrm{SO}(3) \;\to\; n = 8 \text{ symmetry modes } \;\to\; \text{KL-Laplacian has 8 eigenmodes (gap at } \lambda_8)$$

$$\to \text{ nodes partition into 8 firing clusters}$$

Within-cluster: synchrony (positive correlation, shared T_{avg}).
Between-cluster: mass conservation forces anti-correlation.
Net: $\mathrm{Var}(N_i) < \mathrm{Var}_{\text{Poisson}} \;\Rightarrow\; Q_{\text{true}} < 1$.
Correction: $\mathrm{Cov}_{\text{between}}/\mathrm{Var} \approx -0.004 \times 28$ pairs $\;\Rightarrow\; \Delta Q^2 \approx 0.025$, matching the 7% gap.

Why Ψ-field failed as cluster basis. Ψ-field K-means gives $k_{\text{opt}} = 2$ (bimodal $\pm$) because Ψ is an AR(1) diffusion field that has not crystallised into discrete modes at 800 sweeps. The KL-Laplacian encodes the actual firing neighbourhood structure — the level at which $n = 8$ modes live. Ψ is a secondary field; the primary mode structure is in P-space geometry.

7 Status

Result	Status	Note
$Q_{\text{true}} = 0.1613 \pm 0.0023$	Confirmed	5 seeds, CV 1.4%
$Q_{\text{true}} \approx 1/\sqrt{\kappa n T_{\text{avg}}}$	Hypothesis	Step 2 open formally
$k_{\text{opt}} = 8$ (eigengap)	Confirmed	All 5 seeds
H1: $\mathrm{Var}_{\text{within}} < \mathrm{Var}_{\text{global}}$	Confirmed	All 5 seeds
H2: $\mathrm{Cov}_{\text{between}} < 0$	Confirmed	All 5 seeds
$\lambda_2 = (n-1)/T_{\text{avg}}$	Candidate	3.6% gap (4 seeds: 1–2%)
$\tau_{\text{mix}} = T_{\text{avg}}/(n-1)$	Candidate	Physical interpretation
Mechanism: 8 modes + conservation	Hypothesis	Qualitative, 7% match
Formal Step 2 via spectral theory	Open	Priority 1
$Q(\kappa) \propto \kappa^{0.274}$ derivation	Open	Priority 2

8 Open Questions

Priority 1: Formal derivation of λ_2 from δ-chain

Show that the CRF KL-Laplacian has Fiedler value $\lambda_2 = (n-1)^2\varepsilon_0/(2nK^*)$ from the SO(3) symmetry of X-space and the Dirichlet noise ε_0.

Path:

1. Show the KL-graph is approximately n-regular with inter-cluster conductance $h \propto \varepsilon_0/n$.
2. Apply Cheeger inequality: $\lambda_2 \propto h^2$ or the exact spectral gap for stochastic block models with n equal blocks.
3. Connect to Wald formula to obtain the identity.

If proved: formally closes Step 2 and unifies spectral + temporal β-chain.

Priority 2: Derive $Q(\kappa) \propto \kappa^{0.274}$

Path through $\theta_{\text{eff}} = K^*\sqrt{1+\kappa N_{\text{eff}}}$: higher κ raises θ_{eff}, increases T_{avg}, and changes λ_2. Self-consistent loop: $\lambda_2(\kappa) \to T_{\text{avg}}(\kappa) \to Q(\kappa)$. The 0.274 exponent may emerge from this feedback.

The δ-chain chose SO(3). SO(3) chose $n = 8$. The KL-graph encoded 8 modes in its Laplacian spectrum. Each mode fired together. Between modes, the balance was kept. $Q_{\text{true}} < 1$ because the field remembers its own symmetry.

$\lambda_2 = (n-1)/T_{\text{avg}}$*: the mixing time is one mode-crossing. The inter-event time is all of them. The eigengap was always there. It took the right basis to see it.*